Assignment 3: Object Detection Results

Objective. Run object detection on two images and report the class information, confidence, bounding-box coordinates, width, height, area, and centre point.

Implementation. `your-detection.py` loads the Lecture 4 fruit SSD model on the Jetson, detects objects, prints all requested measurements, and draws Type, confidence, L, T, W, H, and Area on the saved output images.

Run command (on Jetson):

```
python3 your-detection.py fruit_2.jpg fruit_5.jpg --model=ssd-mobilenet.onnx --labels=labels.txt
--output-dir=assignment3_results --threshold=0.30
```

Image	Detected class	Confidence	Bounding-box measurements (pixels)
fruit_2.jpg	Strawberry (3 boxes)	31.0%, 48.1%, 40.1%	Full values are printed by the program and drawn on the image.
fruit_5.jpg	Apple; Banana	35.3%; 30.7%	Full values are printed by the program and drawn on the image.

Detailed numerical results

fruit_2.jpg: Strawberry: ClassID 7; confidence 0.3101; Left 506.5; Top 421.8; Right 713.5; Bottom 603.8; Width 207.0; Height 182.1; Area 37690.6; Centre (610.0, 512.8).

fruit_2.jpg: Strawberry: ClassID 7; confidence 0.4815; Left 527.5; Top 115.0; Right 731.0; Bottom 282.9; Width 203.5; Height 167.8; Area 34156.4; Centre (629.2, 199.0).

fruit_2.jpg: Strawberry: ClassID 7; confidence 0.4007; Left 313.2; Top 361.2; Right 500.0; Bottom 490.3; Width 186.8; Height 129.1; Area 24111.5; Centre (406.6, 425.7).

fruit_5.jpg: Apple: ClassID 1; confidence 0.3529; Left 75.8; Top 54.2; Right 588.4; Bottom 972.0; Width 512.6; Height 917.8; Area 470461.6; Centre (332.1, 513.1).

fruit_5.jpg: Banana: ClassID 2; confidence 0.3069; Left 224.1; Top 122.4; Right 543.8; Bottom 419.0; Width 319.7; Height 296.6; Area 94827.3; Centre (383.9, 270.7).

GitHub link: To be added after the code is pushed to the course repository.

Annotated Detection Images



Figure 1. Strawberry detections. Each label contains Type, confidence, L, T, W, H, and Area.

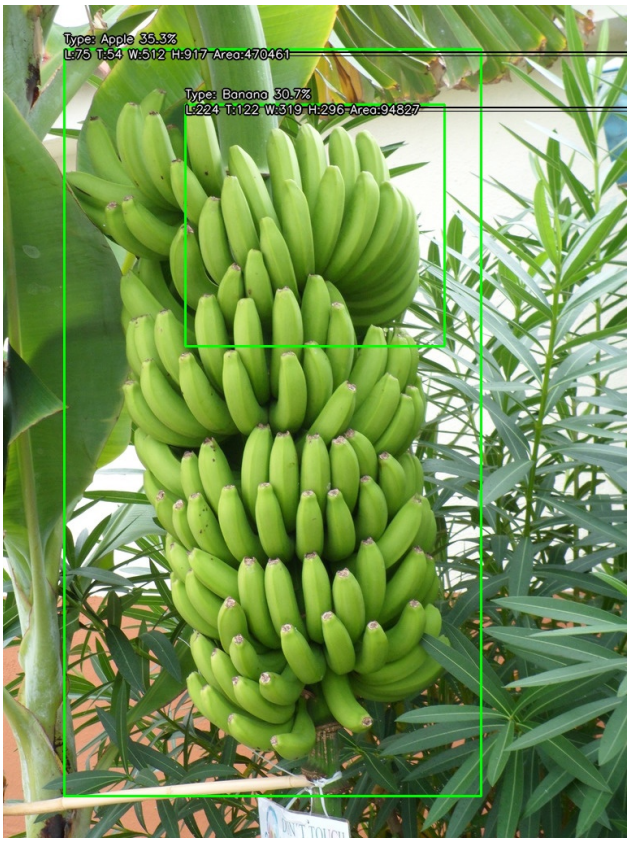


Figure 2. Banana-scene detections. The Apple box is retained as the model output; it demonstrates that the report records actual inference rather than edited labels.